When the State Matches

Justifying Demographic Goals in Dating-App Design

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Slides

Aim and Argument

Aim

Assess Beckstein & de Vries's (2025) defense of demographic weighting without explicit user consent.

Claim

The current defense does not adequately assess whether the weighting's additional benefits justify its effects on encounter opportunities.

My Main Argument

(P1) If the current defense adequately justifies demographic weighting, it provides an adequate assessment of the weighting's additional benefits and the burdens of reduced encounter opportunities.

(P2) The current defense, considered as a whole, does not provide that assessment.

(C) Therefore, the current defense does not adequately justify demographic weighting.

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Two Proposals: Public Apps and Demographic Design

Bouke de Vries (2024)

State-Run Dating Apps: Are They Morally Desirable?

Public provision could correct commercial incentives and help users find lasting relationships.

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Dating Apps as Tools for Social Engineering

Matching design could pursue public goals, including fertility.

Disputed step: Can this justify demographic weighting without explicit user consent?

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de Vries (2024): A Public Dating App

Commercial incentives

Continued searching and spending can generate revenue.

Public provision

Successful relationships can count as success even when users leave the app.

Proposals include priority for users seeking long-term relationships, transparency, independent oversight, and commercial alternatives.

de Vries 2024, pp. 2–3, 9, 12.

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Beckstein and de Vries (2025): Two Designs

	Preference-responsive matching	Demographic weighting
Criterion	Compatible reproductive plans	Expected demographic contribution
Mechanism	User-selected filters	Additional changes to visibility
Purpose	Help users pursue their aims	Advance a population goal

Preference-responsive matching follows users' family plans. Demographic weighting adds a state-selected criterion for how profiles are shown.

Beckstein & de Vries 2025, pp. 5–6, 9.

Beckstein and de Vries: Demographic Weighting

Beckstein and de Vries's proposal (2025)

Favor matches expected to support childbirth or stable child-rearing, even without explicit user consent.

Beckstein & de Vries 2025, p. 9.

Their stated permissibility claim

In some circumstances, the benefits can justify the costs.

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Their Defense: An Argument Reconstruction

(D1) If demographic weighting can secure sufficiently important public benefits while imposing only acceptable burdens, it can be permissible without explicit consent.

(D2) Demographic weighting can secure sufficiently important public benefits while imposing only acceptable burdens.

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Reasons offered for D2

Commercial apps already constrain choices; demographic gains can protect freedom; no relationship is prohibited.

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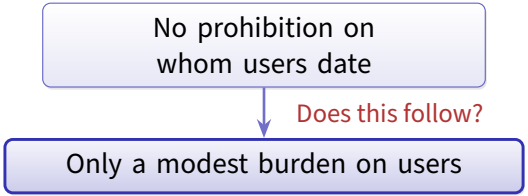
What Their Reasons Establish

Reason offered	What it establishes	Further question
Commercial constraints	Nonintervention preserves an existing design	What does demographic weighting add?
Demographic benefits	A serious public reason	Why use this particular means?
No prohibition	Final choice remains available	How much do encounter opportunities change?

Together, these reasons establish a public purpose and an already structured baseline. They leave open why the additional matching criterion's benefits justify its encounter burdens.

The Inference I Challenge

Beckstein and de Vries (2025, p. 9):



An app can leave every relationship legally available while making some potential partners much harder to encounter. The absence of a ban therefore does not determine how demanding the intervention is.

Beckstein & de Vries 2025, p. 9.

My Counterexample: Reduced Visibility

Suppose

A widely used public app repeatedly reduces some profiles' visibility because of their expected demographic contribution.

(P1) If nonprohibition alone established a modest burden, no intervention could preserve final choice while imposing a substantial burden on encounter opportunities.

(P2) Repeated visibility reductions on a widely relied-on app can preserve final choice while imposing such a burden.

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The Authors' Standard of Freedom

(P1) If effective opportunities matter when assessing an intervention's freedom gains, they also matter when assessing its freedom costs.

(P2) The defense assesses freedom gains by the effective opportunities protected against work and caregiving burdens.

(C) Their assessment of demographic weighting's freedom costs must also consider effective opportunities.

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Star and Ruhleder: Infrastructural Dependence

Encounter opportunities

The significance of a ranking depends on how people rely on the service to meet others.

Star & Ruhleder 1996, pp. 112–113.

- Intensity of the adjustment
- Duration of the adjustment
- Reliance on the service
- Availability of alternatives

My Argument: Public Purposes and Means

(P1) When the state changes citizens' important opportunities for a collective purpose, it must justify that means and its allocation of burdens to the citizens affected.

(P2) Demographic weighting changes citizens' important encounter opportunities for a collective population goal.

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Lazar: Ends, Procedures, and Authority

Ends	Procedures	Authority
What should the system pursue?	How should decisions be made?	Who may exercise this power?

A valuable end leaves questions about both procedure and authority open. The public mandate must cover the criterion through which the system changes citizens' opportunities.

Lazar 2024, preprint, §3.4, pp. 22–24; §4, p. 27.

Three Questions for Demographic Weighting

1. Why pursue population goals through the ordering of intimate encounters?
2. What additional benefits justify the burdens, given feasible alternatives?
3. Who can authorize, challenge, and revise the objective and the weighting?

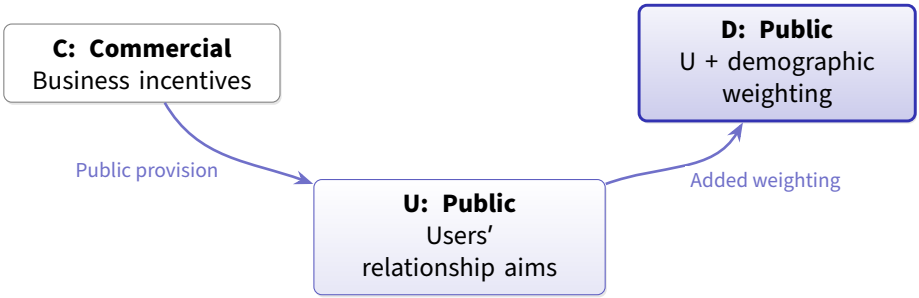
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My Comparison: Public Provision and Weighting



Moving from C to D changes both public provision and demographic direction. If public provision creates gains and weighting removes some of them, D can still outperform C. Comparing D with feasible U isolates the additional weighting's effects.

What the Comparison with C Cannot Establish

(P1) If improvement over C established that weighting improves encounter opportunities, it could not occur when the weighting reduces those opportunities relative to U.

(P2) Such improvement can occur: U can offer the greatest encounter opportunities, D fewer, and C fewer still.

(C) Improvement over C does not establish that the weighting improves encounter opportunities.

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The Combined Defense: The Missing Assessment

Consideration	What it establishes
Nonprohibition	Final choice remains legally available
Commercial replacement	Public provision can improve an existing environment
Demographic benefits	Population goals can supply substantial public reasons
Safeguards	Explanation, monitoring, and contestation can assist assessment

Combining a public goal, a commercial baseline, and review procedures still leaves a question about the intervention itself: why do this weighting's additional gains justify its particular effects on encounter opportunities?

Beckstein & de Vries 2025, pp. 3, 9; de Vries 2024, p. 12; de Vries & González Barman 2026, pp. 320, 330.

My Criticism of the Current Defense

(P1) An adequate assessment must connect the weighting's own additional benefits to reasons for accepting its effects on encounter opportunities.

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Possible Reply: Public Benefit and Freedom

(P1) Every possible design whose public benefits justify its burdens and that is appropriately authorized is permissible.

(P2) Some possible demographic designs have public benefits that justify their burdens and are appropriately authorized.

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(P1) If oversight and explanation can provide the required assessment of benefits and burdens, they can meet the proposed assessment requirement.

(P2) Properly designed oversight and explanation can provide the required assessment of benefits and burdens.

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My Reply: Assessing Benefits and Burdens

(P1) An adequate judgment that benefits justify encounter burdens requires an assessment of the magnitude and distribution of those burdens.

(P2) Nonprohibition and general demographic benefits do not provide that assessment of encounter burdens.

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Urgency, effectiveness, reliance, and feasible alternatives bear on that assessment.

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What can the user change?

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Why is that criterion justified?

What should the state change?

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  Philosophy & Technology 37, article 30.
- Beckstein & de Vries (2025). *Dating Apps as Tools for Social Engineering*.
  Ethics and Information Technology 27, article 11.
- de Vries & González Barman (2026). *Dating Apps and the Right to an Explanation*.
  Journal of Applied Philosophy 43(2): 319–338.

Infrastructure and Governance: References

- Star & Ruhleder (1996). *Steps Toward an Ecology of Infrastructure: Design and Access for Large Information Spaces*.

  Information Systems Research 7(1): 111–134.

- Lazar (2024). *Lecture I: Governing the Algorithmic City*.

  Preprint, arXiv:2410.20720v1.